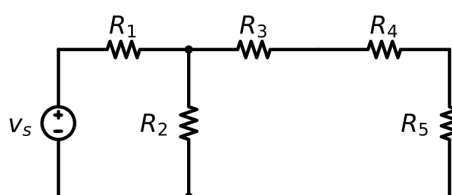


Practice sheet: Chapter 3, Simple Resistive Circuits

This sheet is not marked for correctness, and AI tools are allowed. The device-free quiz at the start of week 5 is built from twins of these questions, so make sure you can do each one on your own.

- **Two-tier items** ask for an answer *and* a reason. Both must be right.
- **Predict first:** where asked, write your prediction before you calculate.
- **Confidence:** next to each answer, write S (sure) or U (unsure). This is not marked; it shows you what to revise.
- **Notation:** $a \parallel b = ab/(a + b)$. Check every answer: resistors in parallel must have the same voltage, and the powers must add to zero.

S1. Series or not? (two-tier)



Which pair of resistors is in series?

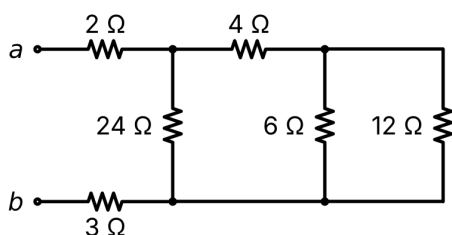
- | | |
|--------------------|--------------------|
| a) R_1 and R_2 | b) R_3 and R_4 |
| c) R_2 and R_5 | d) R_1 and R_3 |

Reason:

- Only those two elements meet at the node between them, so they carry the same current.
- Resistors drawn one after another along a wire are always in series.
- They connect the same two nodes.
- They have the same voltage across them.

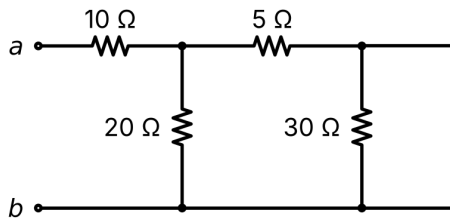
Follow-up: is R_2 in parallel with R_5 ? Explain in one sentence.

S2. Equivalent resistance of a ladder.



Find R_{ab} .

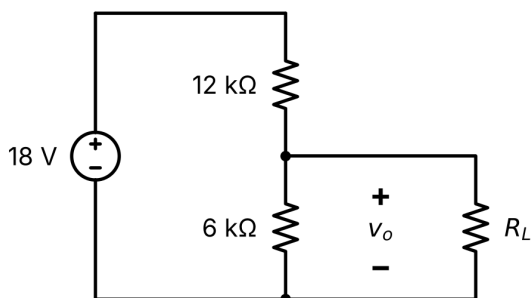
- | | |
|----------------|------------------|
| a) 11 Ω | b) 8 Ω |
| c) 51 Ω | d) 16.5 Ω |

S3. A wire across a resistor (two-tier).Find R_{ab} .

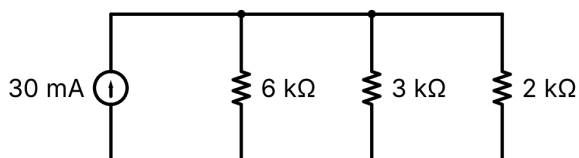
- a) $14\ \Omega$ b) $22.7\ \Omega$
 c) $10\ \Omega$ d) $45\ \Omega$

Reason:

- i) The wire puts 0 V across the $30\ \Omega$ resistor, so it carries no current and the $5\ \Omega$ resistor connects straight to b.
 ii) The wire and the $30\ \Omega$ resistor are in series, so the $30\ \Omega$ stays in the circuit.
 iii) A wire carries no current, so it can be ignored.
 iv) The current splits equally between the wire and the $30\ \Omega$ resistor.

S4. Loading a voltage divider (predict first).

- a) With R_L removed, find v_o .
 b) *Predict first:* when $R_L = 6\ \text{k}\Omega$ is connected, does v_o rise, fall or stay the same? Then find it.
 c) Repeat b) for $R_L = 600\ \text{k}\Omega$. What rule about R_L and the $6\ \text{k}\Omega$ resistor does this suggest?

S5. Current division (two-tier).The current in the $2\ \text{k}\Omega$ resistor is:

- a) $15\ \text{mA}$ b) $5\ \text{mA}$
 c) $10\ \text{mA}$ d) $30\ \text{mA}$

Reason:

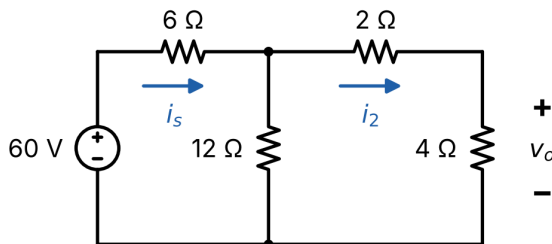
- i) $i_j = (R_{eq}/R_j)i$ with $R_{eq} = 1\ \text{k}\Omega$, so the smallest resistance takes the largest current.
 ii) $i_j = (R_j/R_{eq})i$, so the largest resistance takes the largest current.
 iii) The current splits equally between the three resistors.
 iv) All the current takes the path of least resistance.

Follow-up: find the other two currents and check KCL.

S6. Design a voltage divider. You need 3 V from a 12 V supply, and the divider may draw at most 1 mA from it.

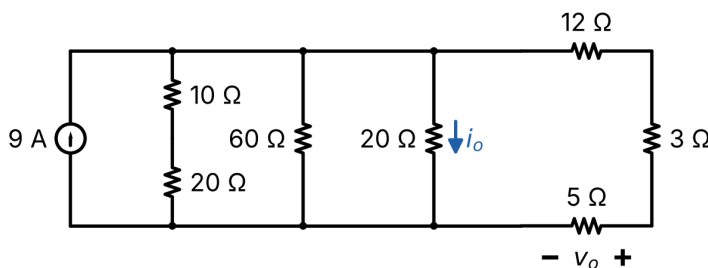
- What is the smallest total resistance $R_1 + R_2$ allowed?
- Choose R_1 (top) and R_2 (bottom).
- The 3 V output now feeds a 3 k Ω load. Find the new output voltage. Is your design still good?

S7. Series-parallel reduction, then division.



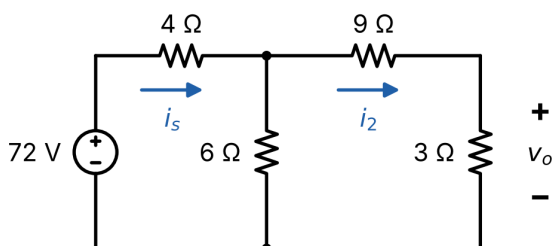
- Find the equivalent resistance seen by the source, and i_s .
- Use current division to find i_2 , then voltage division to find v_o .
- Find the power of every element and check the power balance.

S8. Two divisions in a row. As in Nilsson Example 3.7.

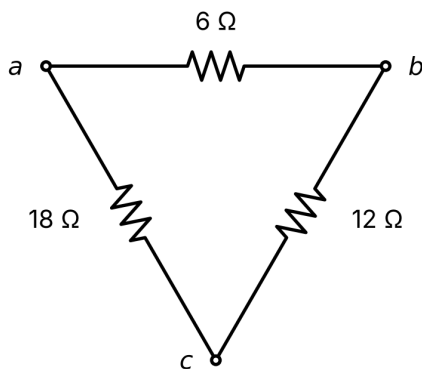


- Find the equivalent resistance seen by the source.
- Use current division to find i_o .
- Use voltage division to find v_o .

S9. Spot the error. The worked solution below finds v_o . It contains the kind of slip AI chat tools often make with the divider rules.



- The 9 Ω and 3 Ω resistors are in series: 12 Ω .
 - $6 \parallel 12 = 72/18 = 4 \Omega$.
 - $R_{eq} = 4 + 4 = 8 \Omega$.
 - $i_s = 72/8 = 9 \text{ A}$.
 - Current division: $i_2 = (12/(6 + 12))(9) = 6 \text{ A}$.
 - $v_o = 3i_2 = 18 \text{ V}$.
 - KCL at the top node: the 6 Ω resistor carries $9 - 6 = 3 \text{ A}$, so the currents balance and the answer is correct.
- Which is the first wrong line?
 - Correct it.
 - Finish the solution and check it.
 - Line 7 says the currents balance. Why is that not a check?

S10. Delta to wye (two-tier).

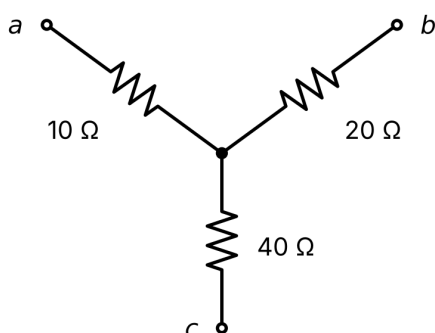
In the equivalent Y, the resistor connected to terminal a is:

- a) $3\ \Omega$ b) $2\ \Omega$
 c) $6\ \Omega$ d) $4.5\ \Omega$

Reason:

- i) Multiply the two Δ resistors that meet at a, and divide by the sum of all three.
- ii) Take the Δ resistor opposite a and divide it by 3.
- iii) Divide the sum of the pairwise products by the Δ resistor opposite a.
- iv) Put the two Δ resistors that meet at a in parallel.

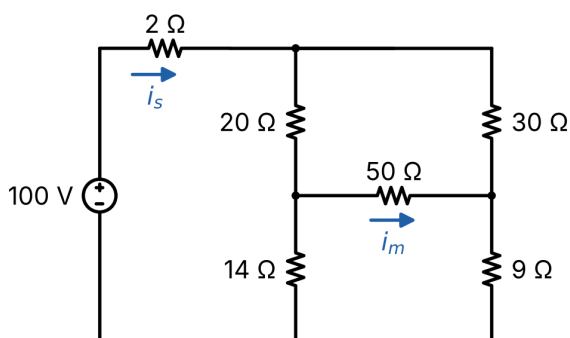
Follow-up: find the other two Y resistors, then check that the resistance between a and b is the same in both circuits.

S11. Wye to delta.

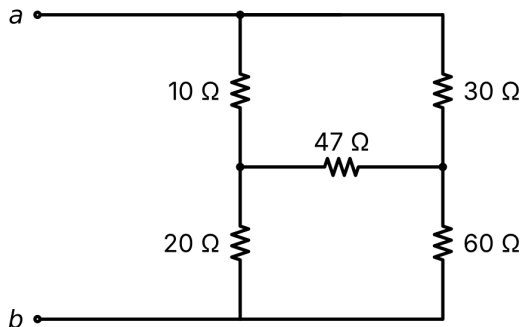
In the equivalent Δ , the resistor between a and b is:

- a) $35\ \Omega$ b) $140\ \Omega$
 c) $70\ \Omega$ d) $30\ \Omega$

Follow-up: find the other two Δ resistors, and check that the resistance between a and b is $30\ \Omega$ in both circuits.

S12. A bridge needs a Δ -to-Y step.

- a) No two resistors here are in series or in parallel. The $20\ \Omega$, $30\ \Omega$ and $50\ \Omega$ resistors form a Δ . Replace it with an equivalent Y: give the Y resistor at the top node, at the left node and at the right node.
- b) Find the equivalent resistance seen by the source, and i_s .
- c) Go back to the original circuit and find i_m in the $50\ \Omega$ resistor.

S13. A balanced bridge (two-tier, predict first).

Predict first: does the 47 Ω resistor change R_{ab} ?
Then find R_{ab} .

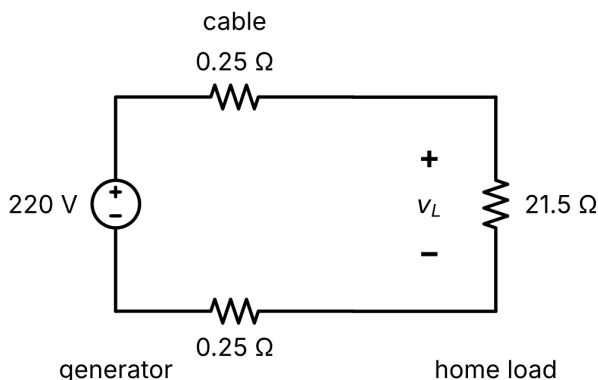
- a) 22.5 Ω b) 15.2 Ω
c) 120 Ω d) 69.5 Ω

Reason:

- i) $10/20 = 30/60$, so both ends of the 47 Ω are at the same voltage and it carries no current.
ii) The 47 Ω is in parallel with the whole network.
iii) The 47 Ω is in series with the whole network.
iv) A resistor across the middle of a bridge never carries current.

S14. Work backwards. In the S7 circuit, replace the 12 Ω resistor with an unknown R . What value of R makes $i_s = 7.5$ A?

- a) 3 Ω b) 2 Ω
c) 6 Ω d) 12 Ω

S15. The cable from your generator (predict first).

A neighbourhood generator holds 220 V at its terminals. Each conductor of the cable to your home has 0.25 Ω , and the home draws power like a 21.5 Ω resistor.

- a) *Predict first:* does the home receive 220 V? Then find the current and v_L .
b) Find the power delivered to the home and the power lost in the cable. What percentage of the generator's output is lost?
c) A thicker cable halves each conductor's resistance. Find v_L now.

S16. How a touch screen finds your finger. In a resistive touch screen (Nilsson's Practical Perspective for this chapter), the grid in the x-direction is a resistance R_x driven by $V_s = 5$ V. A touch splits it into αR_x and $(1 - \alpha)R_x$, and the screen measures V_x across αR_x , so $V_x = \alpha V_s$. The screen is $p_x = 1080$ pixels wide, and the pixel column of the touch is $x = (1 - \alpha)p_x$.

- a) A touch gives $V_x = 2$ V. Find α and the pixel column x .
b) Which voltage-divider idea makes $V_x = \alpha V_s$?
c) Where on the screen is a touch that gives $V_x = 5$ V?